Verifiable Matrix Maintenance with Fresh Work Guarantees

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***Abstract*—Incremental matrix maintenance reuses previous products, but output correctness alone does not quantify the fresh computation behind an accepted update. We derive work-sensitive acceptance bounds for a stateful finite-field interface: a homogeneous bilinear program is selected before fresh update factors arrive, and its complete candidate is fixed before random verification. The processor may retain arbitrary previously paid state. For uniform factors, conditioning on matrix rank yields an output-span agreement profile that improves the corresponding range-size estimate at every positive gate budget below the number of entries in one factor. A resource recursion composes the profiles under adaptive pathwise budgets, including incorrectly accepted intermediate states. Conditional linear escape extends the threshold guarantee to nonuniform factors. Balanced checking minimizes the homogeneous certificates, and an exact allocator minimizes the heterogeneous threshold certificate under equal check costs. A six-product 2×2 Strassen construction attains the single-round threshold bound. Exact-field experiments include candidate materialization and verification: at matrix dimension 4096, update width eight, and field size 65521, reuse reduces graphics-processor latency from 161.93 to 83.16 ms, a paired $1.95\times$ speedup. Randomized categorical views connect report encoding, statistical accuracy, and fresh-work guarantees. At field size 17, parameter-aware checking allocation saves 10.4% of checks relative to fixed-prefix round robin and 1.4% of measured service and planning time over twenty processor sessions.**

I. Introduction

Matrix products arise in analytical views, matrix chains, and feature transformations that must remain current as records change. Incremental view maintenance exploits previous results rather than recomputing them; higher-order maintenance, linear-algebra programs, and factorized analytics provide established techniques for doing so [1]–[3]. Let n be the matrix dimension, r the update width with $n \geq r \geq 2$, and $\mathbb{F}_q$ the finite field with q elements. For $A, B \in \mathbb{F}_q^{n\times n}$ with B fixed and invertible, the maintained product is $C = AB$. Fresh factors $U, V \in \mathbb{F}_q^{n\times r}$ specify $A' = A + UV^\top$, giving the standard update identity

$$C' = A'B = C + U(V^\top B).$$

The identity reduces arithmetic through reuse. Our question is how that reuse interacts with verification when another processor supplies the updated product.

Figure 1 illustrates the motivating analytical view and the causal protocol. In the Covertype application, 40 soil categories and 40 elevation bins of width 100 m over $[0, 4000)$ define the soil–elevation domain. Fresh randomized reports update a retained finite-field matrix; accepted report increments are decoded and debiased into a separate noisy co-occurrence estimate. The heatmap is schematic, not measured data. The latency annotation concerns a distinct $n = 4096$, $r = 8$, $t = 2$, $q = 65521$ dense-GPU experiment, rather than the Covertype view. The two causal boundaries allow reuse of arbitrary paid historical state, while the fresh-work certificate remains conditional on the admitted homogeneous bilinear program class and trusted kernel admission, not physical-work attestation.

Random projections efficiently check a fixed candidate [4], but a correctness check does not by itself measure the computation used to produce it. A processor can retain earlier inputs, products, and auxiliary state, then adapt its computation to the observed history. Relating acceptance to fresh work therefore requires a bound that remains valid after each such history and composes across adaptively chosen updates. Incorrectly accepted intermediate products make this a stateful problem: the next check may start from a reference that is already wrong.

We study a homogeneous bilinear interface in which a program with w gates is fixed before the current factors arrive. Each gate multiplies one linear form in U by one in V; w counts these products across all suppliers. Historical state may select the program, while the complete candidate precedes the verifier's independent random vectors. Section III specifies the interface and its cost accounting. The prototype realizes it through trusted kernel admission.

The key insight is to retain the rank information in the agreement analysis. For independent uniform factors conditional on a fixed history, conditioning on U of rank s makes $UV^\top B$ uniform on an ns-dimensional matrix subspace. The program's outputs lie in the span of its coefficient matrices, whose dimension is at most w. Bounding their intersection within each rank stratum and then averaging yields an explicit agreement profile. It is strictly smaller than the capped range-size estimate for every integer $0 < w < nr$. A backward resource recursion then converts these one-round profiles into a bound for a prescribed sequence under a work budget that holds on every execution path. This separates the effect of retained information from computation on the newly disclosed factors.

Our contributions connect this analysis to checking decisions and maintained analytical state:

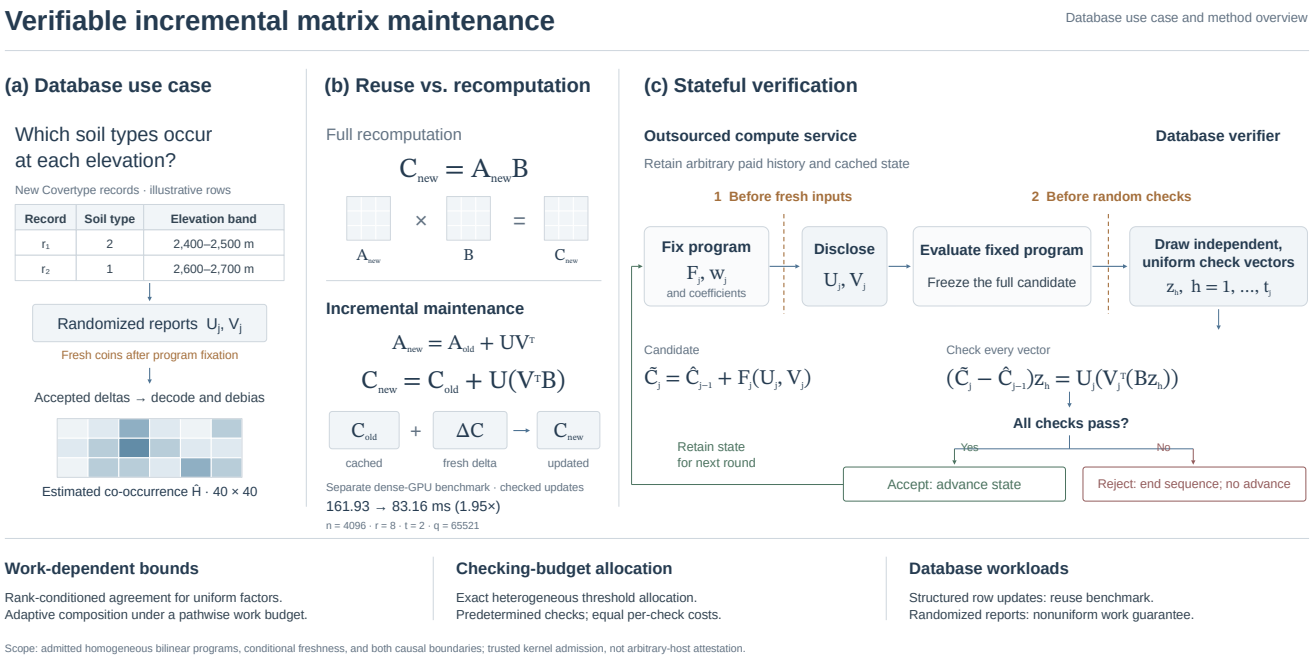


Fig. 1. Database use case and verifiable incremental matrix maintenance. (a) Covertype records yield randomized soil-type and elevation-band reports; the schematic co-occurrence heatmap represents a decoded, debiased estimate, not the retained modular state. (b) A cached product is updated using $U(V^\top B)$ instead of full recomputation; the dense-GPU timing is a separate workload. (c) The admitted bilinear program and gate count are fixed before fresh factors arrive, and the complete candidate is fixed before independent random checks. Acceptance advances the reference; rejection ends the prescribed sequence.

- **Work-sensitive, stateful bounds.** We derive the rank-conditioned agreement profile and its adaptive composition, including incorrect intermediate references (Lemma 3 and Theorem 3). Conditional linear escape supplies threshold bounds for nonuniform factors.
- **Certificate-optimal checking.** We prove balanced allocation optimal for the homogeneous certificates and give an exact allocator for the heterogeneous threshold certificate under equal checking costs (Theorems 5 and 6). A six-product construction attains the single-round threshold bound (Theorem 4).
- **End-to-end evaluation and a nonuniform application.** Exact-field implementations measure reuse together with materialization and checks. Fresh randomized attribute reports instantiate the nonuniform assumptions and expose the relationship between encoding accuracy, work-target feasibility, and integrity-driven checking budgets (Sections V–VI).

II. Related Work

a) Incremental analytics: DBToaster develops higher-order maintenance, LINVIEW treats linear-algebra programs, and F-IVM combines factorization with ring-based analytics [1]–[3]. These techniques optimize maintained values under updates. Our analysis adds a work-sensitive acceptance bound for a processor that retains state, while using the established product update as its application interface.

b) Verification and useful work: Freivalds' random projections verify dense matrix products using quadratic field-operation cost per check [4]. Pantry supports verifiable computation over persistent storage [5], while Dumas and Zucca amortize public matrix-vector verification through preprocessing [6]. These verification mechanisms concern output correctness; our certificates additionally relate acceptance to a fresh bilinear-work budget. Komargodski and Weinstein study useful-work proofs for arbitrary matrix multiplication, with their strongest security claim based on correlated low-rank linear systems [7]. Our guarantees are information-theoretic within the stated online program class. Alman's lookup-table algorithms distinguish finite-field bit complexity from arithmetic-gate counts [8]; our work accounting uses the latter, with other execution costs measured separately.

c) Bilinear agreement: Tensor rank underlies exact bilinear computation [9], [10]. Cohen and Moshkovitz relate analytic, geometric, and slice rank [11], and the geometric rank of rectangular multiplication is known [12]. Using classical finite-field matrix-rank counts [13], our output-span argument conditions on the input rank before averaging the agreement bound. Section IV-D compares the resulting profile with the rank-based and range-size bounds. The six-product witness is related to defective products in probabilistic tensor algorithms [14]; the seven-product threshold also follows from the support-rank result of Bläser et al. [15].

d) Allocation and application mechanisms: Our heterogeneous allocator specializes ordered-sum thresholds [16] and separable concave resource allocation [17] to the acceptance certificate. The categorical application uses established

symmetric and optimized unary encodings [18]: their report distributions determine the conditional escape parameters, connecting encoding accuracy to work-target feasibility. Output integrity imposes a separate checking constraint. Exact word-size field arithmetic follows the implementation approach of FFLAS-FFPACK [19].

III. Model and Stateful Protocol

We study an application that maintains a matrix product as new low-rank updates arrive. The application retains its earlier result; the processor may retain arbitrary paid state. A round has two ordering requirements: the bilinear program is selected before the new factors, and the complete candidate is fixed before the verifier's random vectors (Figure 1).

A. Inputs, state, and computation

Fix a finite field $\mathbb{F}_q$ with q elements, integers $n \geq r \geq 2$, an initial $A_0 \in \mathbb{F}_q^{n\times n}$, and $B \in \mathrm{GL}_n(\mathbb{F}_q)$, the group of invertible $n \times n$ matrices over $\mathbb{F}_q$. The authenticated initial product is $C_0 = A_0B$. Write $\xleftarrow{\$}$ for an independent uniform draw. The intended trajectory of accepted updates then satisfies, in round j,

$$U_j, V_j \xleftarrow{\$} \mathbb{F}_q^{n\times r}, \qquad A_j = A_{j-1} + U_jV_j^\top, \\ C_j = A_jB = C_{j-1} + U_j(V_j^\top B). \tag{1}$$

Uniformity includes rank-deficient factors. Let $\mathcal{H}_j^-$ be the complete classical history just before disclosure: earlier factors, outputs, checking vectors, all suppliers' state, the selected program, its gate allocation, and the coins used to select them. Conditional on every such history, U_j and V_j remain independent and uniform. Section IV-A gives a conditional nonuniform extension.

Preprocessing can depend arbitrarily on A_0, B, and earlier inputs. Its actual cost P is charged, and finite aggregate state S includes all advice, tables, circuits, and caches held by all suppliers. The theorem places no numerical bound on P or S: it constrains the representation and timing of the computation that uses the new factors.

Definition 1 (Admitted online program)**.** *Conditional on $\mathcal{H}_j^-$, the increment is one fixed homogeneous bilinear program: with scalar linear forms $L_{j,a}, R_{j,a}$ in the entries of U, V, output coefficient matrices $Z_{j,a} \in \mathbb{F}_q^{n\times n}$, and gate count w_j,*

$$F_j(U, V) = \sum_{a=1}^{w_j} Z_{j,a}L_{j,a}(U)R_{j,a}(V). \tag{2}$$

The linear forms, output coefficients, topology, and gate count are fixed before the current factors arrive. A gate multiplies one linear form in each fresh factor group. The count includes every supplier's bilinear gates contributing to the round; unused gates may have zero output coefficients.

The fixed transform $V^\top B$ is linear in fresh inputs. Its field operations, state access, additions, communication, and output materialization retain their actual execution costs, but are not the variable-by-variable gates counted by w_j. Exact fast bilinear algorithms are admitted. Input-indexed program selection, branching on current inputs, and nonlinear encodings of those inputs are distinct computational models [8]. Paid state may select the next program before disclosure; an answer computed from the current input by another supplier must be charged as current computation.

B. Candidate and verification

The verifier stores a reference $\widehat{C}_{j-1}$, initially C_0. The candidate is $\widetilde{C}_j = \widehat{C}_{j-1} + F_j(U_j, V_j)$. Its entire actual matrix is fixed before the verifier draws $t_j \geq 1$ independent uniform vectors $z_{j,h} \in \mathbb{F}_q^n$, conditional on the history, factors, and candidate, where $[m] = \{1, \ldots, m\}$ for positive integers m. The verifier tests

$$(\widetilde{C}_j - \widehat{C}_{j-1})z_{j,h} = U_j\big(V_j^\top(Bz_{j,h})\big), \quad h \in [t_j]. \tag{3}$$

Acceptance commits A_j and advances $\widehat{C}_j$ to the candidate. Rejection commits neither update and ends the prescribed sequence. Recovery starts a new sequence from an authenticated, consistent checkpoint, with its own work budget and acceptance event. Failed candidates and transcripts may remain in paid state; freshness in a restarted sequence is conditioned on this full history. The accepted reference alone serves as the previous-product reference. A fully received matrix satisfies the candidate boundary; a cryptographic representation must bind a complete matrix and account for its binding error.

We consider k prescribed attempts with a *pathwise* budget $\sum_{j=1}^{k} w_j \leq W$. Choices may adapt to the complete observed history. Aborting before round k fails the all-execute-and-accept event; selecting successes from unbounded retries is a different event.

A concrete matrix view accumulates cross-products of paired feature streams. A batch of r paired n-dimensional records forms the columns of U_j, V_j; the view maintains $A_j = A_0 + \sum_{i\leq j} U_iV_i^\top$, and fixed B transforms its right-hand features. The uniform model describes independently sampled finite-field records. Independent coordinates with bounded conditional maximum atoms give the nonuniform instance of Corollary 1. These are assumptions on the incoming records, rather than consequences of requesting the product. Section VI also measures a structured row-replacement view through the same update interface.

IV. Fresh Computation and Acceptance Bounds

We first establish the tensor threshold and a history-conditioned single-round bound. Corollary 2 refines this bound as a function of work, and Theorem 3 composes the resulting profiles across adaptive rounds. Theorems 5 and 6 then allocate a fixed budget of checking vectors.

Let $\mathcal{R}_q(n, r, n)$ be the tensor rank of the $\langle n, r, n\rangle$ multiplication map $(U, Y) \mapsto UY$, with $U \in \mathbb{F}_q^{n\times r}$ and $Y \in \mathbb{F}_q^{r\times n}$: the least number of simple terms, each the product of one linear form in U, one linear form in Y, and a fixed output coefficient matrix. Define

$$R = \mathcal{R}_q(n, r, n), \qquad \theta_q = 2/q - 1/q^2 = 1 - (1 - 1/q)^2. \tag{4}$$

Lemma 1 (Fresh-input tensor)**.** *The map $T_B(U,V) = U(V^\top B)$ is linearly isomorphic to the $\langle n,r,n\rangle$ multiplication tensor. Its tensor rank is $R \geq n^2$. For any fixed pre-input state, an admitted program exact on all inputs uses at least R bilinear gates.*

Proof. The input change $\Phi_B(V) = V^\top B$ is invertible, with inverse $\Phi_B^{-1}(Y) = (YB^{-1})^\top$. Tensor rank is invariant under this change, while each summand of (2) is a simple input–input–output tensor. Conditioning on $\mathcal{H}_j^-$ makes all coefficients constants relative to the new inputs, however they were prepared, so the program tensor has rank at most w_j.

For the explicit lower bound, output flattening sends a functional λ to $(U,Y) \mapsto \lambda(UY)$. If this form vanishes, the choices $U = e_i e_1^\top$ and $Y = e_1 e_\ell^\top$, where e_i is the i-th standard basis vector and $E_{i\ell} = e_i e_\ell^\top$ is the corresponding matrix unit, imply $\lambda(E_{i\ell}) = 0$ for every i,ℓ. The flattening is thus injective on an n^2-dimensional space. A simple tensor has flattening rank at most one, proving $R \geq n^2$. These sparse inputs are algebraic witnesses at the same factor width r.

Over a finite field, equality of bilinear functions determines their tensors: evaluation on pairs of input basis vectors recovers every coefficient. Hence $w_j < R$ implies $F_j - T_B$ is a nonzero bilinear map as a tensor and as a function. □

Fast rectangular algorithms can reduce the direct $n^2 r$ gate count, while every exact admitted program still needs at least $R \geq n^2$ gates. This argument counts computation, separately from writing the n^2 output entries.

Theorem 1 (One-round exactness and acceptance)**.** *Fix a history $\mathcal{H}_j^-$ and a program with $w_j < R$. Its increment agrees with T_B with probability at most θ_q. With t checking vectors its acceptance probability is at most*

$$a(t) = \theta_q + (1-\theta_q)q^{-t}. \tag{5}$$

If the preceding reference is genuine, the increment-agreement event is exactly $\widetilde{C}_j = C_j$.

Proof. A nonzero coordinate of $F_j - T_B$ is $p(u,v) = u^\top M v$, where $u = \mathrm{vec}(U)$ and $v = \mathrm{vec}(V)$ stack the entries of U and V in a fixed order, and $\mathrm{rank}(M) = s \geq 1$. Uniformity gives

$$\mathbb{P}[p(u,v) \neq 0 \mid \mathcal{H}_j^-] = (1-q^{-s})(1-q^{-1}) \geq (1-q^{-1})^2. \tag{6}$$

Indeed, $M^\top u \neq 0$ with probability $1 - q^{-s}$; conditional on such a u, its inner product with independent uniform v is uniform in $\mathbb{F}_q$. Full increment agreement implies $p = 0$.

Write $D = F_j(U_j,V_j) - T_B(U_j,V_j)$. The candidate and the checker use the same previous reference, so (3) tests $Dz = 0$ even if that reference is wrong. After candidate fixation, nonzero D passes t independent vectors with probability $q^{-t\,\mathrm{rank}(D)} \leq q^{-t}$ [4]. For increment-agreement probability $p_j \leq \theta_q$, total acceptance is at most $p_j + (1-p_j)q^{-t} \leq a(t)$. □

A. History-conditioned nonuniform factors

Let $u_j = \mathrm{vec}(U_j)$ and $v_j = \mathrm{vec}(V_j)$. For each round, suppose predetermined parameters $\lambda_{U,j}, \lambda_{V,j} \in [0,1]$ satisfy, on every reachable pre-input history h,

$$\mathbb{P}[a^\top u_j \neq 0 \mid h] \geq \lambda_{U,j}, \quad a \neq 0,$$
$$\mathbb{P}[d^\top v_j \neq 0 \mid U_j = u, h] \geq \lambda_{V,j}, \quad d \neq 0,$$

for every supported u. These are uniform lower bounds across histories and nonzero linear forms. The conditioning of the second bound on U_j allows dependencies between the factors without treating them as independent. The invertible change $V \mapsto V^\top B$ preserves the lower bound for nonzero linear forms.

Lemma 2 (Conditional bilinear escape)**.** *Every nonzero bilinear form $u_j^\top M v_j$ is nonzero with conditional probability at least $\lambda_{U,j}\lambda_{V,j}$.*

Proof. Choose a nonzero column a of M. The event $a^\top u_j \neq 0$ has probability at least $\lambda_{U,j}$ and implies $M^\top u_j \neq 0$. For every such realized u_j, the second bound applies to the linear form with coefficient $M^\top u_j$, giving conditional probability at least $\lambda_{V,j}$. Averaging over these u_j proves the claim. □

Corollary 1 (Independent-coordinate escape)**.** *Let $\alpha_j, \beta_j \in [0,1]$. Suppose coordinates of U_j are conditionally independent given h, with maximum atom at most α_j, and coordinates of V_j are conditionally independent given (U_j, h), with maximum atom at most β_j. The escape conditions hold with $\lambda_{U,j} = 1-\alpha_j$ and $\lambda_{V,j} = 1-\beta_j$. For a fixed discrepancy form whose coefficient matrix has rank s, its nonzero probability is at least $(1-\alpha_j^s)(1-\beta_j)$.*

Proof. In a nonzero linear form, condition on all but one coordinate with a nonzero coefficient. At most one value of that coordinate makes the form zero, with conditional probability at most its maximum atom. For the rank-sensitive statement, select s pivot coordinates in the rank-s system $M^\top u_j = 0$. Conditional on the remaining coordinates, the pivot values are uniquely determined. Their conditional independence bounds this event by α_j^s. On its complement, apply the single-form bound $1-\beta_j$ conditional on U_j. □

Uniform coordinates give escape $1 - 1/q$; independent Bernoulli(p) coordinates give at least $\min\{p, 1-p\}$. Conversely, a factor supported on a linear hyperplane can have zero escape. Structured data streams must establish the stated conditional parameters before these probability guarantees apply; measuring an update operator's runtime does not supply those distributional parameters.

B. Heterogeneous adaptive acceptance

Theorem 2 (Stateful aggregate bound)**.** *For the prescribed k rounds, assume the conditional escape bounds of Lemma 2 hold on every reachable pre-input history, including histories with incorrectly accepted updates. Let $t_j \geq 1$ be predetermined and the check vectors follow* (3)*. The escape parameters are predetermined history-uniform bounds as defined*

in Section IV-A, not history-varying realizations. Suppose $\sum_j w_j \le W$ *on every path. Put* $b = \min\{k, \lfloor W/R \rfloor\}$, $\theta_j = 1 - \lambda_{U,j}\lambda_{V,j}$ *and* $a_j = \theta_j + (1-\theta_j)q^{-t_j}$. *Then*

$$\mathbb{P}[\textit{all } k \textit{ rounds execute and accept}] \le \max_{\substack{E\subseteq[k]\\ |E|\le b}} \prod_{j\notin E} a_j. \tag{7}$$

The programs and allocations may adapt to the complete observed history; an abort or rejection fails this event. The same maximum with θ_j *in place of* a_j *bounds the probability that all* k *rounds execute and their increments are exact; an abort before round* k *fails this event.*

Proof. Write $T_j = U_j V_j^\top B$ and $D_j = F_j(U_j, V_j) - T_j$. On every realized pre-input history, F_j is fixed. If $w_j < R$, Lemma 1 gives a nonzero bilinear discrepancy coordinate, so Lemma 2 gives $\mathbb{P}[D_j = 0 \mid \mathcal{H}_j^-] \le \theta_j$. This argument does not require a genuine stored reference. The check subtracts the same reference used to form the candidate, and is exactly $D_j z_{j,h} = 0$. Given the factors and frozen candidate, a nonzero D_j passes all independent vectors with probability $q^{-t_j \operatorname{rank}(D_j)} \le q^{-t_j}$; thus a cheap round accepts with conditional probability at most a_j. An expensive round ($w_j \ge R$) accepts with probability at most one.

At most b rounds are expensive on any path. For s remaining expensive allowances, set $V_{k+1}(s) = 1$ and recursively define $V_j(s) = \max\{a_j V_{j+1}(s), V_{j+1}(s-1)\}$, omitting the second term when $s = 0$. Backward induction over complete histories bounds the remaining all-accept probability by $V_j(s)$: the cheap branch uses its conditional acceptance bound and the worst continuation after each accepted history; the expensive branch spends one allowance. Randomized choices are averaged, and aborts cannot increase success. Expanding the recurrence proves (7); replacing acceptance by $D_j = 0$ proves the increment statement. No independence across rounds is needed. □

a) Conditional protocol composition: An implementation transfers the theorem's bound when it enforces the selected program, its evaluation relation, the shared previous-state reference, the pathwise allocation, and the two causal boundaries. If its execution can be coupled to the ideal process except with joint probability ϵ over the prescribed sequence, the all-accept bound increases by at most ϵ. This follows by a union bound without conditioning freshness on absence of failure. Our local candidate digest fixes bytes before check generation; it is not a public attestation of program execution or physical work. With $E_j = \widehat{C}_j - C_j$ and $D_j = F_j(U_j, V_j) - U_j V_j^\top B$, accepted states obey $E_j = E_{j-1} + D_j$: an exact increment preserves an existing state error, and later errors can cancel.

C. Uniform factors and singular fixed operands

For uniform factors and $t_j = t$, put $b = \min\{k, \lfloor W/R \rfloor\}$. Theorem 2 gives

$$\mathbb{P}[\text{all } k \text{ attempts execute and accept}] \le a(t)^{k-b}. \tag{8}$$

This holds for adaptive programs and allocations under the pathwise budget. It is one when $b = k$. The work-sensitive profile below retains the individual gate counts instead of only this expensive-round count.

If $\operatorname{rank}(B) = s$, write $B = P \operatorname{diag}(I_s, 0) Q$ with invertible P, Q, where I_s is the $s \times s$ identity matrix. The input change $V' = P^\top V$ and output change $C \mapsto CQ^{-1}$ give $(U, V') \mapsto [U(V'_{1:s,:})^\top \ 0]$, where $V'_{1:s,:}$ denotes the first s rows of V'. Restriction and projection recover $\langle n, r, s\rangle$, and embedding gives the reverse reduction. The fresh tensor rank is therefore $\mathcal{R}_q(n, r, s) \ge ns$, by the same output flattening. At $B = 0$ the increment is zero. The main workload uses invertible B.

D. A Work-Sensitive Finite-Field Bound

Use the interface of Section III with $n \ge r \ge 2$, invertible B, and conditionally independent uniform $U, V \in \mathbb{F}_q^{n\times r}$.

Lemma 3 (Output-span agreement)**.** *For* $0 \le s \le r$, *let* ρ_s *be the probability that a uniform* $n \times r$ *matrix over* $\mathbb{F}_q$ *has rank* s, *and define*

$$\rho_s = q^{-nr} \prod_{i=0}^{s-1} \frac{(q^n - q^i)(q^r - q^i)}{q^s - q^i}, \tag{9}$$

with empty product one, and put

$$\mu_{n,r,q}(w) = \sum_{s=0}^{r} \rho_s q^{-(ns-w)_+}, \qquad (x)_+ = \max\{x, 0\}. \tag{10}$$

For every reachable pre-input history $\mathcal{H}^-$ *and admitted* w*-gate program* F,

$$\Pr[F(U, V) = UV^\top B \mid \mathcal{H}^-] \le \mu_{n,r,q}(w), \tag{11}$$

with equality when $w = 0$.

Proof. The outputs of $F = \sum_{a=1}^{w} Z_a L_a(U) R_a(V)$ belong to the fixed space $Z = \operatorname{span}\{Z_1, \dots, Z_w\}$, of dimension at most w. Given rank-s U, the independent uniform matrix $Y = V^\top B$ makes UY uniform on its ns-dimensional image $\mathcal{L}_U$. Thus

$$\begin{aligned}\Pr[F = UY \mid U, \mathcal{H}^-] &\le \Pr[UY \in Z \mid U, \mathcal{H}^-] \\ &= q^{\dim(Z\cap\mathcal{L}_U) - ns} \le q^{-(ns-w)_+}.\end{aligned} \tag{12}$$

Counting full-rank $n \times s$ and $s \times r$ factorizations and dividing by their $|\mathrm{GL}_s(\mathbb{F}_q)|$ multiplicity gives the rank probabilities (9) [13]. Averaging proves the claim, including zero and rank-deficient inputs. At $w = 0$, $F = 0$, and conditional agreement has probability q^{-ns}. □

Write $\mu(w) = \mu_{n,r,q}(w)$ and $\vartheta = 2/q - 1/q^2$.

a) Comparison with bilinear agreement bounds: For this comparison let $T(U, V) = UV^\top B$ be the fresh target and $D = T - F$ the discrepancy of an admitted program. Write $\mathrm{AR}(D)$, $\mathrm{GR}(D)$, and $\mathrm{SR}(D)$ for the analytic rank, the geometric rank, and the slice rank of D: AR is $-\log_q \Pr[D = 0]$ for uniform bilinear inputs, GR is the codimension of the zero set over the algebraic closure, and SR is the least

number of slices, each a linear form in one mode times a bilinear form in the other two. Cohen and Moshkovitz (CM) establish $\mathrm{AR}(D) \geq (1 - \log_q 2)\,\mathrm{GR}(D)$ [11, Proposition 4.1 and Fact 4.5]. Rectangular multiplication has geometric rank $g = nr - \lfloor r^2/4 \rfloor$ [12] and slice rank nr even over the algebraic closure, by its semistability [20, Example 24 and Lemma 23]; semistability here means that the zero tensor is not in the Zariski closure of the orbit under determinant-one basis changes in each mode.[1] Take slice rank over the algebraic closure. Subadditivity gives $\mathrm{SR}(D) \geq nr - w$ and $\mathrm{GR}(D) \geq g - w$. Cohen–Moshkovitz's proof gives $\mathrm{SR}(D) \leq 2\,\mathrm{GR}(D)$ over that closure [11, proof of Theorem 3.1]; no descent is needed. Since geometric rank is integral, their finite-field analytic-rank inequality yields

$$\Pr[F = T] \leq (2/q)^{\max\{\lceil (nr-w)/2 \rceil,\, g-w,\, 0\}}, \quad q > 2. \tag{13}$$

An elementary comparison also uses the exact zero probability: a uniform-input bilinear map's largest output atom occurs at zero [11, proof of Proposition 5.1]. The fixed range of F has at most q^w points, giving $B_{\mathrm{range}}(w) = \min\{1, q^w \mu(0)\}$. Concavity of $z \mapsto \min\{1, z\}$ gives

$$\mu(w) = \mathbb{E}_s \min\{1, q^{w-ns}\} \leq B_{\mathrm{range}}(w). \tag{14}$$

This is strict for every integer $0 < w < nr$: the positive rank-zero mass makes $\mu(w) < q^w \mu(0)$, and positive full-rank mass makes $\mu(w) < 1$. At $w = 0$ the two range bounds coincide exactly. At $n = 16, r = 4, q = 17, w = 56$, $\mu(w) = 1.434 \times 10^{-10}$, $B_{\mathrm{range}}(w) = 1.426 \times 10^{-5}$, and (13) gives 1.916×10^{-4}. The rank-conditioned argument also covers $q = 2$, where the geometric comparison's coefficient vanishes. Below a certified tensor-rank threshold, all three bounds can also use the common cap $2/q - 1/q^2$; checking then applies $p \mapsto p + (1-p)q^{-t}$.

Choose a certified integer lower bound $1 \leq R_0 \leq R = \mathcal{R}_q(n, r, n)$; Lemma 1 permits $R_0 = n^2$, without asserting equality to R. Define

$$\overline{p}(w) = \begin{cases} \min\{\vartheta, \mu(w)\}, & 0 \leq w < R_0, \\ 1, & w \geq R_0, \end{cases} \tag{15}$$

and, for predetermined t_j,

$$\mathsf{Acc}_j(w) = \overline{p}(w) + (1 - \overline{p}(w)) q^{-t_j}. \tag{16}$$

Corollary 2 (A conditional acceptance profile)**.** *At every reachable pre-input history, exact-increment and acceptance probabilities are at most $\overline{p}(w)$ and $\mathsf{Acc}_j(w)$, respectively, even with an incorrect previous reference.*

Proof. For $w < R_0 \leq R$, a nonzero bilinear coordinate of the discrepancy vanishes with probability at most ϑ by Theorem 1. Combining Lemma 3 gives (15); for $w \geq R_0$ the bound is trivial. The candidate and checker use the same old reference, leaving discrepancy $D = F(U,V) - UV^\top B$ regardless of prior errors. A fixed nonzero D passes all checks with probability $q^{-t_j \operatorname{rank} D} \leq q^{-t_j}$. Conditioning on $D = 0$ proves (16). □

Increment and complete-product agreement coincide only with a genuine reference. Acceptance can be strictly smaller even at $w = 0$ when the nonzero discrepancy has rank above one. Transforms, state access, materialization, and checking retain their separate costs.

E. Adaptive Rounds Under a Pathwise Work Budget

The recursion below is a backward dynamic program (DP) over the remaining work budget.

Theorem 3 (Aggregate work-profile bound)**.** *Consider k prescribed rounds under the preceding conditional freshness assumptions, predetermined $t_1, \ldots, t_k$, and $\sum_j w_j \leq W$ on every path. Put $\overline{W} = \lfloor W \rfloor$ and $G_{k+1}(z) = 1$ for integer $z \geq 0$. Define backwards*

$$G_j(z) = \max_{\substack{w \in \mathbb{Z}_{\geq 0} \\ w \leq z}} \mathsf{Acc}_j(w) G_{j+1}(z - w). \tag{17}$$

Then

$$\begin{aligned} &\Pr[\textit{all } k \textit{ rounds execute and accept}] \leq G_1(\overline{W}) \\ &\quad = \max_{\substack{\mathbf{w} \in \mathbb{Z}_{\geq 0}^k \\ \sum_j w_j \leq \overline{W}}} \prod_{j=1}^{k} \mathsf{Acc}_j(w_j). \end{aligned} \tag{18}$$

Replacing $\mathsf{Acc}_j(w)$ by $\overline{p}(w)$ bounds the event that all k rounds execute with exact increments.

Proof. At a pre-input history with remaining budget z and selected allocation w, Corollary 2 bounds acceptance by $\mathsf{Acc}_j(w)$. Every accepted continuation has budget at most $z - w$; backward induction bounds its success by $G_{j+1}(z-w)$ uniformly over complete histories, including incorrect references. Maximizing over w proves (17); randomized choices are mixtures. Aborts and rejections fail the event. Expanding the recurrence proves (18); replacing acceptance by exactness proves the final statement. Cross-round independence is unnecessary. □

a) Finite actions and the threshold comparison: Since $\mu(w) = 1$ for $w \geq nr$, the actions reduce to

$$\mathcal{A} = \{0, \ldots, \min(nr, R_0 - 1)\} \cup \{R_0\}, \tag{19}$$

intersected with the remaining budget. If $nr < R_0$, retain nr as the least-cost representative of the $\overline{p}(w) = \vartheta$ plateau through $R_0 - 1$; R_0 represents all larger actions. These replacements preserve profile values at no greater cost. The bound is one for $\overline{W} \geq kR_0$. For the default $R_0 = n^2$, improvement over the threshold profile requires $w < nr$, or normalized work $w/R_0 < r/n$.

With acceptance coefficients precomputed and $M = |\mathcal{A}| \leq \min(nr, R_0 - 1) + 2$, value computation takes $O(k(\overline{W}+1)M)$ arithmetic operations/comparisons and $O(\overline{W}+1)$ auxiliary DP

[1] If a slice decomposition uses dimensions a_i in modes of dimensions m_i with $\sum_i a_i < \min_i m_i$, assign weights $1 - a_i/m_i$ to its subspaces and $-a_i/m_i$ to their complements. After clearing denominators, each factor has total weight zero, while every supported tensor term has positive weight $\geq 1 - \sum_i a_i/m_i$. This contradicts semistability. The reverse bound follows by slicing a smallest mode.

storage. Recovering an allocation stores one maximizing action per state, requiring $O(k(\overline{W}+1))$ additional entries. These are arithmetic/comparison-model counts. Replacing $\overline{p}(w)$ below R_0 by ϑ recovers the threshold bound: free below-threshold rounds and at most $\min\{k, \lfloor W/R_0 \rfloor\}$ threshold-reaching rounds. The profile strengthens it pointwise.

b) An analytical example: For $n = 4, r = 2, q = 2, k = 4, W = 8, t_j = 1, R_0 = 16$, the 256 matrices U have rank counts $(1, 45, 210)$. Hence

$$\mu(w) = \frac{1 + 45\,2^{-(4-w)_+} + 210\,2^{-(8-w)_+}}{256}. \tag{20}$$

Here $\mu(0) = 593/32768$, $\mu(7) = 151/256 < \vartheta = 3/4$, and $\mu(8) = 1$, so action 8 is essential. A maximizing allocation is $(8, 0, 0, 0)$ up to permutation, giving

$$\begin{aligned} G_1(8) &= \frac{7}{8}\left(\frac{33361}{65536}\right)^3 \approx 0.115421, \\ \text{threshold bound} &= \left(\frac{7}{8}\right)^4 \approx 0.586182. \end{aligned} \tag{21}$$

Indeed, $\log \mathsf{Acc}(w) = \log(A + B2^w)$ is convex on $[0, 4]$ for positive A, B, concentrating work at the endpoints. At most one allocation exceeds four; comparing $(4, 4), (5, 3), (6, 2), (7, 1), (8, 0)$ with the other two allocations zero establishes the maximum.

F. Single-Round Sharpness

Karppa and Kaski's six-product Strassen–Winograd variant omits one of the seven bilinear products [14, Sec. 1.1]. A witness from Strassen's identity [10] attains our one-round bound.

Theorem 4 (Sharpness of the one-round threshold)**.** *For every finite field $\mathbb{F}_q$, $t \geq 1$, $n = r = 2$, and invertible B, an admitted six-gate program has exact-increment probability $\vartheta = 2/q - 1/q^2$ and acceptance probability exactly $\vartheta + (1 - \vartheta)q^{-t}$. Its six gates are fewer than the exact threshold $\mathcal{R}_q(2, 2, 2) = 7$.*

Proof. Write $U = \left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$, $Y = V^\top B = \left[\begin{smallmatrix} e & f \\ g & h \end{smallmatrix}\right]$. Compute

$$\begin{aligned} P_1 &= (a + d)(e + h), & P_2 &= (c + d)e, \\ P_3 &= a(f - h), & P_4 &= d(g - e), \\ P_5 &= (a + b)h, & P_7 &= (b - d)(g + h), \end{aligned}$$

and return

$$F_6 = \begin{bmatrix} P_1 + P_4 - P_5 + P_7 & P_3 + P_5 \\ P_2 + P_4 & P_1 - P_2 + P_3 \end{bmatrix}.$$

The seven-product identity gives

$$F_6 - UY = -E_{22}(c - a)(e + f). \tag{22}$$

The omitted product is never computed. Its two linear forms are independent uniform field elements, including in characteristic two, so their product vanishes with probability $1-(1-1/q)^2 = \vartheta$. Otherwise the discrepancy has rank one and passes t independent checks with probability exactly q^{-t}. The invertible input change preserves the six-gate representation; $\mathcal{R}_q(2, 2, 2) = 7$ over every field [15]. □

The existing missing-term construction therefore establishes single-round sharpness.

G. Selecting and Distributing Checking Work

Checking counts are predetermined and equally costly; programs may still adapt to prior history.

Corollary 3 (A closed-form checking rule)**.** *Put $b = \min\{k, \lfloor W/R_0 \rfloor\}$, $d = k - b > 0$, and $\alpha(t) = \vartheta + (1 - \vartheta)q^{-t}$. For $\delta \in (0, 1)$, the uniform threshold certificate $\alpha(t)^d \leq \delta$ holds exactly when*

$$d \geq \left\lceil \frac{\log \delta}{\log \alpha(t)} \right\rceil.$$

For fixed d, a finite t exists exactly when $\delta > \vartheta^d$; the smallest permitted integer is

$$t_{\min} = \max\left\{1, \left\lceil \log_q \frac{1 - \vartheta}{\delta^{1/d} - \vartheta} \right\rceil\right\}. \tag{23}$$

Proof. The equivalent inequality $q^{-t} \leq (\delta^{1/d} - \vartheta)/(1 - \vartheta)$ requires a positive right side for finite t. Taking logarithms gives both expressions. □

For $d = 0$ the certificate is one. The floor ϑ^d concerns this threshold certificate; use (17) for the stronger profile.

Theorem 5 (Balanced checking for identical rounds)**.** *Assume identical $n, r, q, \overline{p}$ and equal checking costs. Among integer $t_j \geq 1$ with $\sum_j t_j = T \geq k$, a balanced allocation minimizes both the threshold and work-profile certificates. Write $T = k\ell + s$, $0 \leq s < k$. A minimizer has $k - s$ entries ℓ and s entries $\ell + 1$; its threshold value, for $d = k - b$, is*

$$\alpha(\ell)^{\min\{d, k-s\}}\, \alpha(\ell + 1)^{\max\{0, d-k+s\}}. \tag{24}$$

Its work-profile value is given by (17)*.*

Proof. For $c \in [0, 1]$, $f_c(x) = \log[c + (1 - c)q^{-x}]$ is convex. Thus

$$H(\mathbf{t}) = \max_{\substack{\mathbf{w} \in \mathbb{Z}^k_{\geq 0} \\ \sum_j w_j \leq \lfloor W \rfloor}} \sum_j f_{\overline{p}(w_j)}(t_j)$$

is convex; identical profiles and a permutation-invariant budget make it symmetric. The log threshold certificate, a maximum of sums of d functions $f_\vartheta(t_j)$, has the same properties. For integers $u - v \geq 2$, replacing (u, v) by $(u - 1, v + 1)$ averages the original pair with its swap, with swap weight $1/(u - v)$. Convexity and symmetry cannot increase either objective. Iteration reaches the stated balanced allocation. Its d largest $\alpha(t_j)$ contain $\min\{d, k - s\}$ copies of $\alpha(\ell)$, proving the formula. □

If $s \leq b$, the exceptional set contains all strengthened rounds and the threshold value remains $\alpha(\ell)^d$; the profile still uses its recurrence.

H. Heterogeneous Checking Allocation

For the predetermined history-uniform parameters of Theorem 2, put $\vartheta_j = 1 - \lambda_{U,j}\lambda_{V,j}$ and

$$\alpha_j(t) = \vartheta_j + (1-\vartheta_j)q^{-t}, \qquad c_j(t) = -\log \alpha_j(t).$$

For a valid bound $b < k$ on threshold-reaching rounds and $d = k - b$, optimize

$$\Psi(\mathbf{t}) = \max_{\substack{E\subseteq[k]\\|E|\le b}} \prod_{j\notin E} \alpha_j(t_j) \tag{25}$$

over integer $t_j \ge 1$, $\sum_j t_j = T \ge k$. We specialize ordered-sum thresholds [16] and greedy concave resource allocation [17].

Theorem 6 (An exact heterogeneous checking allocation)**.** *Under equal checking costs and the preceding parameters, form*

$$\mathcal{C} = \{c_j(u) : 1 \le j \le k,\ 1 \le u \le T-k+1\}.$$

For each $\tau \in \mathcal{C}$, start with $t_j = 1$ and repeatedly allocate the remaining $T-k$ checks to a round maximizing

$$\Delta_j = \min\{c_j(t_j+1), \tau\} - \min\{c_j(t_j), \tau\}, \tag{26}$$

breaking ties by fixed index. Select the candidate and threshold maximizing

$$\sum_j \min\{c_j(t_j), \tau\} - b\tau. \tag{27}$$

The selected allocation globally minimizes Ψ at budget T.

Proof. Since $0 < \alpha_j \le 1$, $-\log\Psi$ is the sum of the d smallest $c_j(t_j)$. In nondecreasing order $c_{(i)}$, the ordered-sum identity is

$$\sum_{i=1}^{d} c_{(i)} = \max_{\tau\ge 0}\left\{\sum_j \min(c_j, \tau) - b\tau\right\}. \tag{28}$$

Its slope is the number of entries above τ minus b; hence $\tau = c_{(d)}$ maximizes it, including ties and $b = 0$. For fixed τ, $\min\{c_j(t), \tau\}$ is nondecreasing and concave. Its discrete gains are nonnegative and nonincreasing. Each feasible allocation takes a prefix of every gain sequence. Taking the largest available gain preserves these prefixes and selects the largest remaining gains, proving fixed-threshold greedy optimality. Every feasible $t_j \le T-k+1$, so an optimal allocation's threshold $c_{(d)}$ belongs to $\mathcal{C}$. Interchanging the two maximizations and enumerating that set therefore attains the global optimum. □

Put $S = T - k$. At most $k(S+1)$ thresholds each require S allocations. A heap uses $O(k(S+1)(k + S\log k))$ reward comparisons in the exact-comparison model, giving pseudo-polynomial complexity. Parameters must remain uniform over all reachable histories, including incorrect acceptances. Optimality concerns (25), not the heterogeneous work-profile minimax objective or actual attack probability.

a) Local descent and integrity requirements: Directly maximizing the one-step decrease of Ψ need not be globally optimal. For $k = 3, b = 2, q = 17, \vartheta_j = 15/16, T = 6$, all available gains are zero along the fixed-smallest-index greedy path from $(1,1,1)$. It returns $(4,1,1)$ with $\Psi = 16/17$, whereas $(2,2,2)$ gives $271/289 < 16/17$.

Conditional checking soundness and a union bound limit the probability of any incorrect accepted increment by $I(\mathbf{t}) = \sum_j q^{-t_j}$, without independence across rounds. For $0 < \vartheta_j \le 1$, define

$$f_\infty = \max_{\substack{E\subseteq[k]\\|E|\le b}} \prod_{j\notin E} \vartheta_j, \qquad \kappa = \max_j \frac{1-\vartheta_j}{\vartheta_j}.$$

Then

$$f_\infty \le \Psi(\mathbf{t}) \le f_\infty \exp\big(\kappa I(\mathbf{t})\big). \tag{29}$$

Indeed, $\alpha_j(t) = \vartheta_j[1 + (1-\vartheta_j)q^{-t}/\vartheta_j]$; apply $1 + x \le e^x$ to each retained factor and maximize over E.

Theorem 6 also holds with integer minima $t_j \ge L \ge 1$ and $T \ge kL$: start from L, allocate $T - kL$ checks, and enumerate

$$\mathcal{C}_L = \{c_j(u) : j \in [k],\ u = L, \dots, T-(k-1)L\}.$$

The same threshold and decreasing-gain proof gives global optimality among allocations obeying these minima. For $0 < \varepsilon < 1$, choosing $L = \max\{1, \lceil \log_q(k/\varepsilon)\rceil\}$ ensures $I(\mathbf{t}) \le \varepsilon$; this common minimum is a sufficient condition for a general integrity budget.

Figure 2 compares the analytical bounds and the checking allocations: (a) $n = 16$, $r = 4$, $q = 17$, with $40 \le w \le 64$ and the common cap ϑ since $w \le 64 < R_0 = n^2$; (b) $n = 8$, $r = 2$, $q = 17$, $k = 4$, $t = 2$, $R_0 = 64$; (c) $q = 17$, $k = 512$, $t_j \ge 1$, $W \le 128R_0$, alternating $\vartheta_j = 15/16$ and $8/9$, with the exact allocation (OPT) and fixed-order round robin (RR). The dotted target is $(31/50)^{64}$. Panel (a) is a single-round agreement bound; panels (b) and (c) are aggregate all-accept bounds under the work-budget hypothesis.

V. Execution and Resource Costs

A. Platforms and matched operators

The primary server (fy) has four RTX 5090 graphics processing units (GPUs). Four DGX Spark hosts (Spark1–Spark4) each provide a GB10 GPU and a 20-core Arm central processing unit (CPU); the host Spark-ebaf (ebaf for short) supplies the Arm CPU measurements. Python 3.12.3, NumPy 2.5.4, and CuPy 14.2.0 support the word-size backend; Basic Linear Algebra Subprograms (BLAS) on the CPU use four threads. The large-field C++17 kernel is single-threaded. The machines are shared and retain their existing workloads. Independent sweeps use separate devices; each host's long-stream extension runs sequentially.

The incremental operator computes $Y = V^\top B$ and UY, then adds the retained product. Full recomputation forms $A'B$. Both pipelines maintain $A' = A + UV^\top$ when the application retains A. Table I separates this common update from product construction. These direct counts also distinguish the work model's fresh bilinear products from fixed-right transforms,

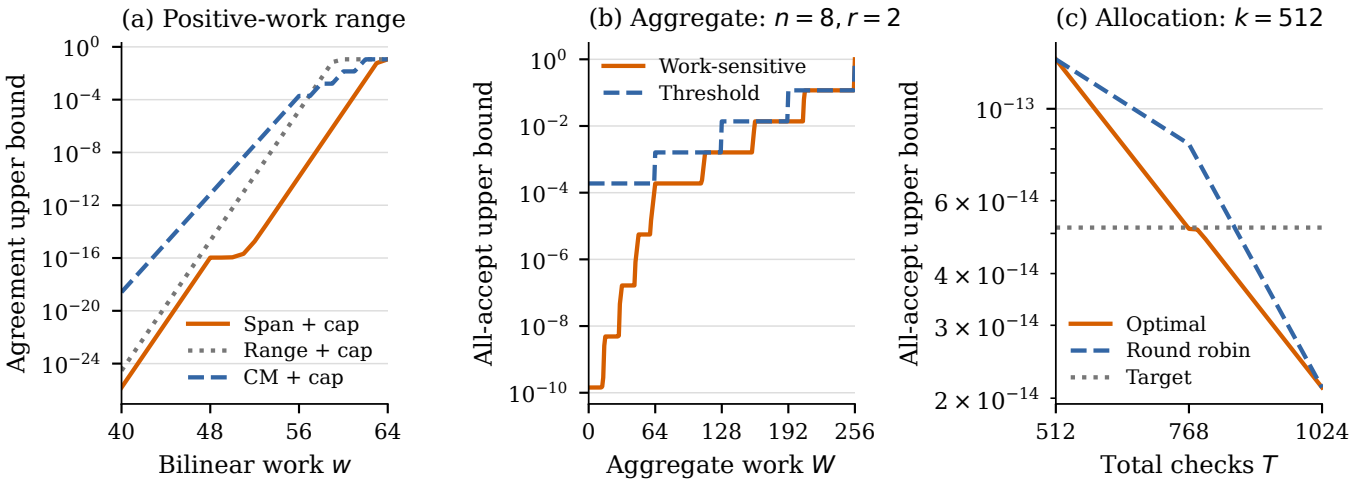


Fig. 2. Analytical agreement and all-accept upper bounds on logarithmic probability axes.

TABLE I
DIRECT PER-ROUND MULTIPLICATION COUNTS FOR FULL RECOMPUTATION, INCREMENTAL MAINTENANCE, AND VERIFICATION.

Operation	Multiplications
Common $UV^\top$	$n^2 r$
Increment $V^\top B$, then $U(V^\top B)$	$2n^2 r$
Full $A'B$	n^3
One incremental check vector	$2n^2 + 2nr$

verification, and state access. The theoretical model admits fast bilinear programs.

B. Exact arithmetic and candidate interface

The main backend uses $q = 65521$ with canonical residues. Double-precision floating-point (FP64) general matrix multiplication (GEMM) evaluates each product, followed by conversion to signed 64-bit integers and modular reduction. Every tested inner dimension $d \le 4096$ satisfies

$$d(q-1)^2 \le 4096 \cdot 65520^2 < 2^{44} < 2^{53}.$$

All nonnegative integer products and partial sums are exactly representable; subsequent additions are integral. This follows the word-size-field approach of exact linear algebra [19]. The separate $q = 2^{61} - 1$ backend uses 128-bit products, reducing every product and accumulated sum modulo q.

The NumPy/CuPy backends materialize a complete canonical 32-bit candidate, copy it into a read-only host snapshot, and hash its bytes with the round context before generating checks. The $q = 2^{61} - 1$ backend instead delivers a separate immutable 64-bit matrix without a digest. The received copy is checked using operating-system randomness and unbiased rejection sampling. Rejection leaves the reference unchanged and ends the sequence. The digest records byte fixation; the program-class guarantee uses the execution contract of Section IV-B. Retaining A, B, C requires $3n^2$ field words. At $n = 4096$ the candidate alone occupies 64 MiB; the $r = 8$ GPU run's maximum live-array allocation is 1.75 GB, including benchmark reset snapshots.

C. Structured analytical views

Let $X, Y \in \mathbb{F}_q^{n \times d}$ contain n records with d features, with Y fixed. Maintain $A = XY^\top$ and $C = AB$. Replacing b rows of X with difference $D \in \mathbb{F}_q^{b \times d}$ and selector $E \in \mathbb{F}_q^{n \times b}$ gives

$$\Delta A = E(DY^\top), \qquad \Delta C = E((DY^\top)B). \tag{30}$$

Implementations use row gather/scatter. A factorized baseline caches $H = Y^\top B$ and applies $E(DH)$, charging construction of H and its dn retained field words. Setup-inclusive comparisons use each method's actual initialization, including XH where applicable. The deterministic selector has zero linear escape, so this view measures structured reuse. The randomized-report view below also instantiates the nonuniform work guarantee.

Covertype contains $581,012$ records and 54 integer features [21]. The row-replacement study excludes the class label and maps features into $\mathbb{F}_{65521}$. Five disjoint source regions provide initial tables, reference tables, and unrecycled replacement streams, using seed identifiers $101, 211, 307, 401, 503$. Each trial uses n rows per initial table and kb replacement rows. Methods share inputs and rotate execution order across seed/repeat pairs.

D. Randomized categorical views

We maintain a cross-product of reusable attribute reports for Covertype's 40 soil categories and 100-meter elevation bins in $[0, 4000)$ ($n = 40$). Each arriving record supplies one paired report; the aggregate is retained across rounds. Symmetric unary encoding (SUE) reports a true bit with probability $p_j = 1 - \eta_j$ and a false bit with probability $\beta_j = \eta_j \in \{1/4, 1/3\}$. For a fresh-work instance, all report coins are independent and sampled after program fixation.

Conditioning on arbitrary source records and history leaves coordinate atoms at most $1-\eta_j$, hence $\lambda_U = \lambda_V = \eta_j$. Each attribute has privacy budget $\varepsilon_j = 2\log((1-\eta_j)/\eta_j)$; a paired report has $2\varepsilon_j$. Privacy-matched optimized unary encoding (OUE) uses $p_j = 1/2$ and $\beta_j \in \{1/10, 1/5\}$ [18], giving escape $\lambda_U = \lambda_V = \beta_j$.

For a prime $q > 2$, set $r = 2$, $B = I$, and retain one modular state C, where I is the identity matrix, $\mathbf{1}_r$ and $\mathbf{1}$ are all-ones vectors of length r and of the ambient dimension, and lift_q maps a field element to its least nonnegative integer representative. The verifier computes $m_U = U_j\mathbf{1}_r$ and $m_V = V_j\mathbf{1}_r$ from received reports. Correct increments lie in $[0,2]$, so their unique lift, the decoding increment $\Gamma_j = \mathrm{lift}_q(\widetilde{C}_j - \widehat{C}_{j-1})$, recovers $U_jV_j^\top$ despite cumulative wrap. An increment entry outside $\{0,1,2\}$ causes rejection without state advance. On acceptance, advance C and accumulate outside the field:

$$\widehat{H} = \sum_j \frac{\Gamma_j - \beta_j m_U \mathbf{1}^\top - \beta_j \mathbf{1} m_V^\top + r\beta_j^2 \mathbf{1}\mathbf{1}^\top}{(p_j - \beta_j)^2}.$$

Correct maintenance gives an unbiased estimator $\widehat{H}$ of the joint category counts by independence of the two report noises. The uniform-factor work profile is unnecessary for this nonuniform application.

Alternate the two parameter types over 512 rounds. Under the global budget $W \le 128n^2$, $R \ge n^2$ permits the conservative allowance $b = 128$ in Theorem 2, using the largest 384 acceptance coefficients. Sorting all 512 rounds gives the global work-certificate floors

$$f_{\mathrm{SUE}} = (625/1024)^{64}, \qquad f_{\mathrm{OUE}} = \big((99/100)^4(24/25)^2\big)^{64}.$$

SUE can certify the target $(31/50)^{64}$; OUE cannot reach it through this threshold certificate.

For the two alternating types, the exact allocator admits a faster specialization of (25). Let $m = 256$, $b = 128$, integer $T \ge 2m$, $S = T - 2m$, and give S_A extra checks to the $\eta = 1/4$ type and $S - S_A$ to the $\eta = 1/3$ type. Convexity and permutation symmetry within each type, as in Theorem 5, ensure a within-type balanced optimum. Because $15/16 > 8/9 + (1/9)q^{-1}$ for $q \ge 17$, the objective always retains all m first-type coefficients and the largest $m - b$ second-type coefficients. Its logarithm is affine between the integer breakpoints

$$\{0, S\} \cup \big(\{hm,\ S - hm,\ S - hm - b : h \in \mathbb{Z}_{\ge 0}\} \cap [0, S]\big).$$

At a multiple of m a balanced level changes; the shifted breakpoints occur when the b excluded second-type entries cross a level. Evaluating these endpoints with exact rational products therefore finds the global optimum at fixed T. Monotonicity in T then permits a search for the smallest total meeting the target, verified against the optimum at the preceding integer total.

Section VI-D combines the work and integrity requirements at $q = 65521$; the $q = 17$ configuration isolates work-target allocation. Estimation error remains a separate statistical quantity. The implemented fixed bilinear evaluator uses trusted kernel admission and input disclosure; its gate count is a logical resource measure, not hostile-host attestation.

E. Measurement

Matrix-engine latency starts with fresh-input transfer and ends after the accepted state advances. It includes candidate copies, applicable hashing, challenge generation, checks, and necessary device synchronization. Fixtures, resets, and correctness oracles are recorded separately. The unchecked ablation removes only checking from the same candidate interface. Synchronized component diagnostics use separate runs and are not substituted for end-to-end latency.

A complete update includes common maintenance, transfers, freezing, enabled checking, and state advancement. Each matrix-engine round is compared with a full product on the same backend; small products and selected coordinates are independently checked with Python integers. A mismatch stops that run. Live device arrays, reserved allocator memory, reset snapshots, and process-lifetime resident set size (RSS) are distinguished. The randomized-report service validates typed-plan admission, candidate fixation, reject-without-advance, and per-round integer decoding; a post-stream integer reconstruction checks its complete decoded output. Its verification model and statistical noise are reported separately.

VI. Experimental Evaluation

A. Design and statistical units

The matrix-engine evaluation contains a 73-cell grid and six additional long-stream runs: $176,520$ method-round records, with no full-product discrepancy. Each cell uses five workload seeds and three repeats. Per-update estimates take medians over rounds, repeats, then seeds; speedups use within-seed ratios. Error bars are nominal 95% percentile-bootstrap intervals from $2,000$ resamples of the five seed summaries. Their small-sample resolution and shared-machine timing limit population-level interpretation. Cumulative bands are pointwise and include initialization. The extension reuses the same seeds and retains its separate estimates. The connected study uses twenty same-tape pairs for $q = 17$ replay, twenty fresh source-paired streams per encoding for $q = 65521$ service, and twenty independent utility simulations, displaying the individual units in Figure 5.

The fixed Covertype extension uses $b = 8, t = 2, k = 256$ at $n \in \{256, 512, 1024\}$ on server GPU2 and $n \in \{512, 1024\}$ on a Spark CPU, plus one synchronized $n = 1024$ diagnostic. Jobs run sequentially within each host. The artifact preserves the full grid, source identities, seed summaries, setup costs, and timing boundaries. The code and lightweight reproducibility materials are available at [2].

B. Strength of the bounds

Figure 2(a) compares the work profile with both the combined slice/geometric-rank and exact-zero-probability range bounds. The positive budget $w = 56$ example improves the

[2]https://github.com/fuyuefyu/verifiable-matrix-maintenance

latter by approximately 9.95×10^4, locating the gain in rank conditioning rather than zero-work agreement. Composition also changes the certificate: at $n = 4, r = 2, q = 2, k = 4, W = 8, t = 1$, the refined bound is 0.115421 versus the threshold bound 0.586182, even though no round can reach $R_0 = 16$.

Exact enumeration validates the rank formula on 417 matrices, profile allocation on $7,294$ comparisons, and balanced allocation on 360 comparisons. The heterogeneous allocator agrees across $15,147$ configurations and $1,651,995$ candidate comparisons, including $3,360$ cases with tied optima. Its grid covers $q \in \{2, 5, 17\}$, $k = 2, \ldots, 5$, $T = k, \ldots, k + 8$, and $b = 0, \ldots, k - 1$, with uniform, mixed, and endpoint profiles. The single-round witness is checked on $794,884$ input pairs and $19,651,396$ explicit checking vectors over fields of size two, three, and five. These are evaluations of mathematical bounds and procedures, not measured attack probabilities.

C. Complete-update benefits and bottlenecks

At $n = 4096, r = 8, t = 2$, checked GPU maintenance takes 83.162 ms against 161.931 ms for checked recomputation, with paired speedup $1.947\times$ ($[1.931, 1.985]$). The $r = 32$ sweep reaches $1.968\times$ ($[1.913, 1.986]$). At $n = 1024$ and 2048, the $r = 8$ gains are $1.167\times$ and $1.522\times$; the $n = 512$ interval $[0.959, 1.105]$ includes parity. Figure 3 uses $q = 65521$ and $k = 8$: (a) complete-update latency on fy GPU0 with $r = 8$ and $t = 2$; (b) latency on fy GPU1 with $n = 1024$ and $t = 2$, tested widths at equal horizontal spacing; (c) within-seed checked/unchecked ratios on fy GPU1 with $n = 1024$ and $r = 8$. It also shows the update-width crossover: at $n = 1024$, speedup falls from $1.256\times$ for $r = 2$ to $0.997\times$ for $r = 512$ ($[0.986, 1.026]$). The complete pipeline's checking factors for $t = 1, 2, 4, 8$ are $1.130, 1.101, 1.099, 1.111$; batched kernels and shared candidate costs prevent a linear extrapolation in t.

The structured Covertype view provides complementary reuse (Figure 4). That figure uses 54 features over $\mathbb{F}_{65521}$ at $n = 1024$, $b = 8$, and $t = 2$: (a,b) accumulate 256 updates on fy GPU2 and the Spark-ebaf CPU from round-zero initialization with applicable cache preparation; Full, incremental, and cached denote recomputation, incremental maintenance, and cached-factorized maintenance; (c) is a separate eight-update synchronized GPU diagnostic whose product, freezing, and checking medians are not additive. At $n = 1024, b = 8$, GPU incremental latency is 4.463 ms versus 5.712 ms, a paired $1.279\times$ gain ($[1.227, 1.280]$). Setup-inclusive medians over 256 updates are 1750.736, 1330.161, and 1325.996 ms for recomputation, generic maintenance, and cached factorization. Caching adds little to the generic operator: its paired factor is 1.001 ($[0.995, 1.002]$). The batch-width sweep at $b = 2, 8, 32$ likewise places all extra-cache intervals across parity.

Synchronized diagnostics explain this result. At the same view size, full product construction takes 1.483 ms, versus 0.0702 ms for the generic increment and 0.0588 ms with caching. Generic candidate freezing takes 3.032 ms and checking 0.441 ms. Retaining the product removes a substantial stage; the additional cache optimizes a much smaller stage behind a common materialization cost. Separately summarized stage medians are not added to construct a total. Dense diagnostics show the same cost structure.

Smaller views locate the useful regime: GPU speedups are 0.932 at $n = 256$ and 0.982 at 512. At $n = 512$ the new stream totals overlap broadly; the earlier collection's generic total was 807.668 ms versus 608.057 ms for recomputation. Stream length alone does not remove costs paid for every candidate. These complete collections remain in the artifact.

The CPU extension at $n = 1024$ takes 10.234 versus 21.468 ms, giving $2.098\times$ ($[2.086, 2.112]$). Its setup-inclusive totals are 2645.888 versus 5520.294 ms; caching gives a further $1.020\times$ ($[1.010, 1.029]$). Matched cached checking factors are 2.953 on CPU and 1.110 on GPU. Across the four GB10 devices, dense $n = 1024, r = 8$ speedups range from 1.225 to 1.925; one interval spans $[1.147, 4.656]$, so devices are not pooled. The separate $q = 2^{61} - 1$ integer backend reaches $20.714\times$ ($[17.998, 20.970]$) at $n = 256, r = 2$, and $8.699\times$ ($[8.506, 8.752]$) at $r = 8$. Those subsets use different Spark hosts and support within-cell comparisons.

D. Encoding and checking choices for randomized views

The categorical view connects a report-encoding choice to an accuracy requirement and a fresh-work target on a 40×40 domain, with privacy-matched SUE and OUE. Utility uses twenty independently simulated randomizations at four nested source prefixes, with common random numbers between the privacy-matched encodings within a replicate. The performance comparison uses twenty shared report tapes per policy; the live-service runs instead generate new operating-system report coins after each fixed program. Replaying a tape contributes no new fresh-work episode. Utility simulations and small-field replays are distinct from the fresh-input integrity-service runs.

a) Statistical accuracy: Figure 5(a,b) evaluates all 1,600 cells and the twenty cells selected by the true full-data counts before examining noisy estimates; all estimates are unclipped. With N the number of source records and H the true count table, frequency mean squared error (MSE) is $n^{-2}\|\widehat{H}/N - H/N\|_F^2$, with $\|\cdot\|_F$ the Frobenius norm. Panel (a) places the four nested source prefixes at equal horizontal spacing, shows all twenty replicate frequency root mean squared error (RMSE) values, and draws solid curves as the root mean MSE, the square root of the mean replicate MSE, and dashed curves as $\sqrt{\mathbb{E}[\mathrm{MSE}]}$. For a single-parameter stratum, write p and β for the common p_j and β_j, and put $\sigma_0^2 = \beta(1-\beta)/(p-\beta)^2$ and $\sigma_1^2 = p(1-p)/(p-\beta)^2$. Independence of attribute noises makes the sum of cell variances per record $[(n-1)\sigma_0^2 + \sigma_1^2 + 1]^2 - 1$, regardless of its categories. For the equal parameter strata, the whole-table frequency MSE therefore satisfies

$$\mathbb{E}[\mathrm{MSE}]_{\mathrm{SUE}} = \frac{47}{20N}, \qquad \mathbb{E}[\mathrm{MSE}]_{\mathrm{OUE}} = \frac{1925689}{1036800N}.$$

Panel (b) draws one line per replicate pairing SUE and OUE at $N = 581{,}012$ on the twenty cells selected by the true full-data counts, with fixed tie breaking, and marks the

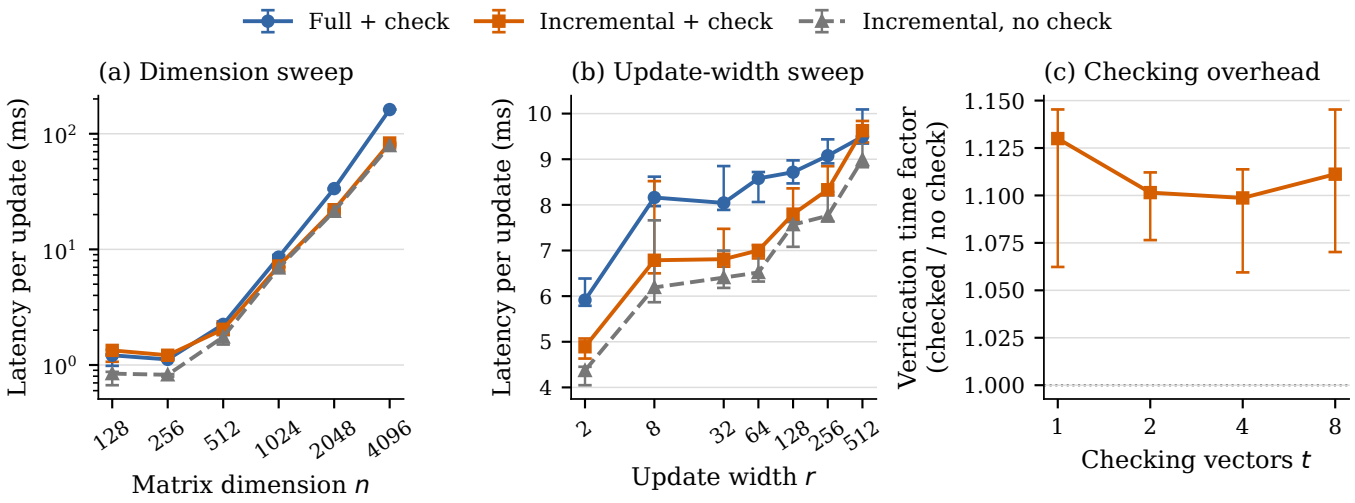


Fig. 3. End-to-end matrix-maintenance latency and checking overhead across matrix sizes and update widths.

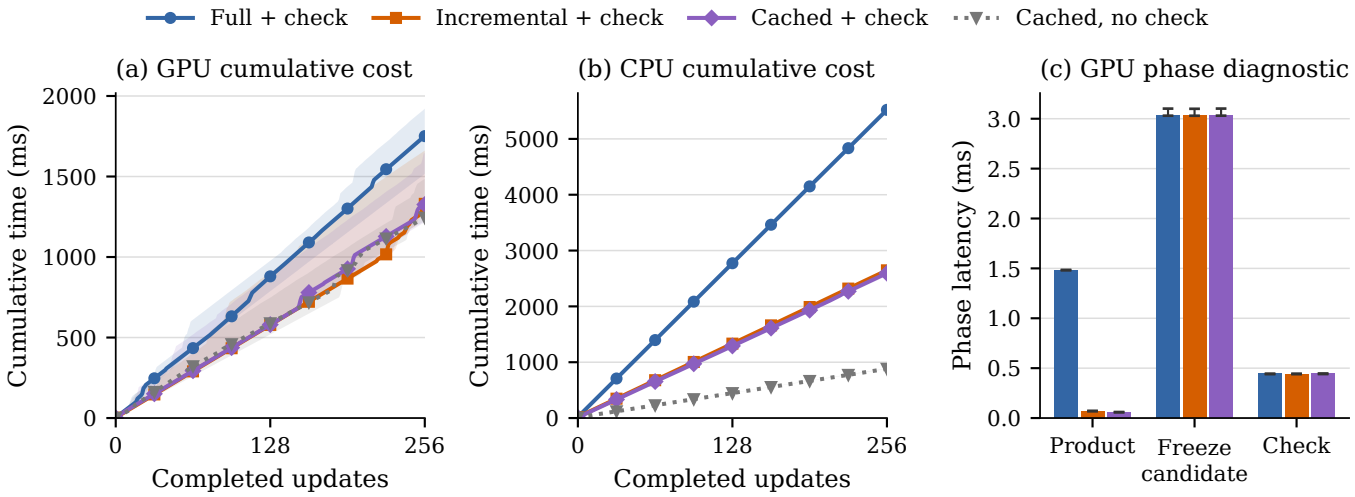


Fig. 4. Setup-inclusive latency and component costs for the Covertype row-replacement view.

median frequency RMSE in black. Across all four prefixes and both encodings, empirical mean MSE is within 0.8% of its analytical value. At $N = 581{,}012$, the whole-table root mean MSE is 2.018×10^{-3} for SUE and 1.781×10^{-3} for OUE: their ratio is 1.133 with nominal interval $[1.123, 1.142]$. The median paired top-twenty RMSE ratio is 1.138 ($[1.046, 1.357]$). Thus SUE's stronger work certificate in this application accompanies about 13% higher whole-table root mean MSE; the two encoding choices expose a quantifiable accuracy–work-guarantee tradeoff.

b) From a target to measured checking cost: At $q = 17$, with $k = 512$ and $r = 2$, the exact allocator (OPT) certifies the minimum of 767 checks for $\delta = (31/50)^{64} = 5.165 \times 10^{-14}$. Certificate-marginal greedy starts with $t_j = 1$ and repeatedly maximizes $\log \Psi(\mathbf{t}) - \log \Psi(\mathbf{t} + \mathbf{e}_j)$, where $\mathbf{e}_j$ is the j-th standard unit vector, breaking ties by fixed index. It reaches the same 767-check schedule here; fixed-prefix round robin (RR) requires 856. Adapting checks to the round parameters therefore saves 10.4% relative to RR. The preceding budgets 766 and 855 fail the corresponding exact target tests. Theorem 6 provides the general global-optimality guarantee. The specialized procedure agrees with exhaustive enumeration on 162 small cases (6, 804 allocations) and 56 full-size comparisons.

Observed complete plan-generation charges are 3.473 and 2.743 ms for OPT and RR. A separate matched search-only comparison on fy, using five blocks with rotating policy order, gives medians 1.553, 0.866, and 34.443 ms for OPT, RR, and certificate-marginal greedy, using cached coefficients and incremental certificate evaluation. These spans exclude serialization and output validation; the service comparison below retains the complete plan-generation charges.

We charge the complete fy plan-generation cost once per twenty-session batch. Service execution runs separately on the

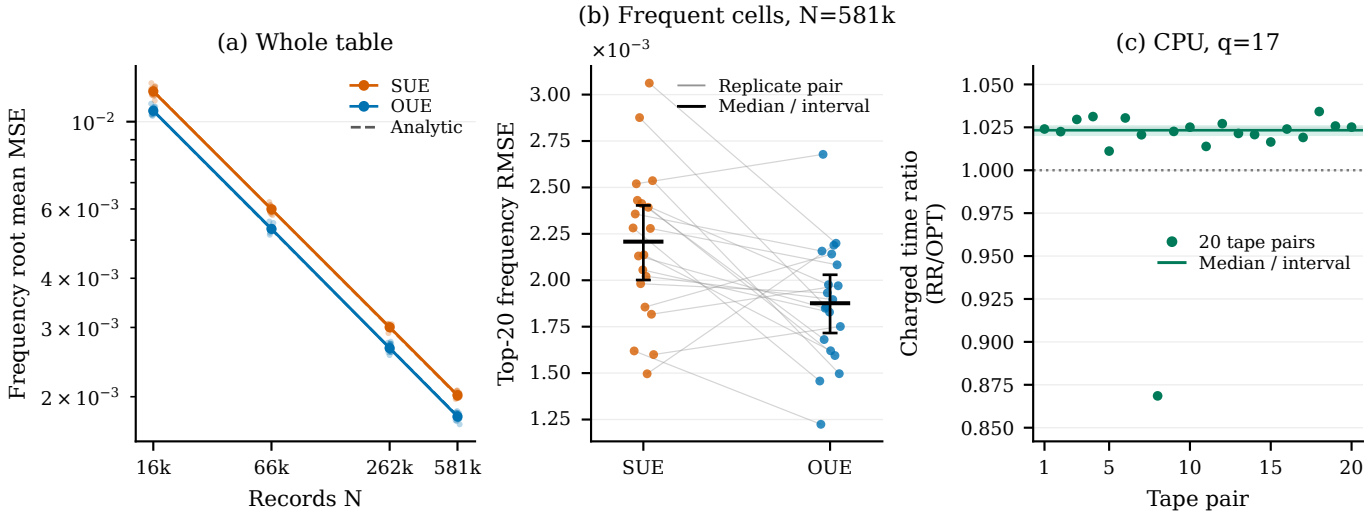


Fig. 5. Statistical accuracy and checking costs for maintained Covertype categorical views.

ebaf CPU; each stream includes plan reading, initialization, complete candidates, sequential checks, and decoding. Figure 5(c) uses twenty CPU replay pairs from this service and allocates the measured plan-generation charge equally across sessions. For policy $a \in \{\mathrm{RR}, \mathrm{OPT}\}$, let τ_a denote measured service time, and let π_a denote the plan-generation time; the charged ratio is $(\tau_{\mathrm{RR}} + \pi_{\mathrm{RR}}/20)/(\tau_{\mathrm{OPT}} + \pi_{\mathrm{OPT}}/20)$. Reported intervals are nominal 95% paired percentile-bootstrap intervals over twenty replicate or stream IDs within each experiment; the intervals in (a) preserve the nested prefixes and are pointwise. The median paired ratio is 1.0233 with nominal interval $[1.0207, 1.0254]$; the twenty-session service-plus-planning totals are 997.488 ms for RR and 983.342 ms for OPT. Nineteen of twenty CPU pairs favor OPT; the remaining ratio is 0.869, so the full-batch saving is 1.4%. The corresponding GPU paired ratio is 1.0459 ($[1.0417, 1.0590]$), and the batch ratio is 1.0482. These intervals condition on the observed planning cost and shared-machine schedule. These $q = 17$ policies are matched on the work target; their output-integrity certificates differ.

c) Fresh-input service at the integrity setting: At $q = 65521$, impose the common integrity certificate $I(\mathbf{t}) \le \varepsilon = 512/q^2 = 1.193 \times 10^{-7}$. Since $q > 512$, one round with $t_j = 1$ already exceeds this budget. Thus every round needs at least two checks and $T \ge 1024$. For SUE, $\kappa = 1/8$ and $f_{\mathrm{SUE}} \exp(\varepsilon/8) < \delta$; by (29), uniform $t = 2$ attains the minimum $T = 1024$ under both certificates. Its work bound is 1.893×10^{-14}. OUE's floor $4.105 \times 10^{-4} > \delta$ precludes this work target through the threshold certificate. Here integrity determines the checking count, while encoding determines work-target feasibility. These are analytical certificates under the stated hypotheses, not measured attack rates. Table II reports twenty fresh streams per encoding and backend on the Spark-ebaf CPU and fy's RTX 5090 GPU0, with $q = 65521$ and median milliseconds per 512-round stream. Live time is the continuous interval including report generation; active time removes measured client-generation spans and is a discontinuous subtotal. All 160 replay/live streams complete their 512 rounds and their final decoded arrays match direct integer reconstruction from the reports. At this 40×40 size, the tested ebaf CPU is the lower-latency choice than fy's RTX 5090 GPU0; GPU per-round transfers and kernel launches remain visible in the complete candidate interface.

TABLE II
MEDIAN LIVE AND ACTIVE SERVICE LATENCY IN MILLISECONDS FOR 512-ROUND FRESH-INPUT STREAMS.

Backend	Encoding	Live time	Active time
CPU	SUE	65.289	51.773
CPU	OUE	64.445	51.592
GPU	SUE	627.120	589.921
GPU	OUE	624.246	588.208

VII. Discussion and Conclusion

Retained matrix products and arbitrary paid history can coexist with acceptance bounds under a fresh-work budget. Conditioning on rank strengthens agreement at positive work budgets, and a resource recursion composes the per-round bounds under adaptive pathwise spending, including rounds that accept an already incorrect reference. Checking allocation then turns an acceptance target into a finite resource decision, kept separate from the integrity budget that governs output correctness. Exact-field measurements locate the scale at which arithmetic reuse outweighs candidate materialization, and fresh randomized attribute reports connect encoding, statistical accuracy, and verification cost. These guarantees hold for the stated causal ordering and bilinear program class. The prototype exercises that interface through trusted kernel admission; accounting for the same work under untrusted general-purpose execution would require a separate enforcement mechanism.

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References

[1] Y. Ahmad, O. Kennedy, C. Koch, and M. Nikolic, "DBToaster: Higher-order delta processing for dynamic, frequently fresh views," *Proceedings of the VLDB Endowment*, vol. 5, no. 10, pp. 968–979, 2012.

[2] M. Nikolic, M. ElSeidy, and C. Koch, "LINVIEW: Incremental view maintenance for complex analytical queries," in *Proceedings of the 2014 ACM SIGMOD International Conference on Management of Data*. ACM, 2014, pp. 253–264.

[3] A. Kara, M. Nikolic, D. Olteanu, and H. Zhang, "F-IVM: Analytics over relational databases under updates," *The VLDB Journal*, vol. 33, pp. 903–929, 2024.

[4] R. Freivalds, "Fast probabilistic algorithms," in *Mathematical Foundations of Computer Science 1979*, ser. Lecture Notes in Computer Science, vol. 74. Springer, 1979, pp. 57–69.

[5] B. Braun, A. J. Feldman, Z. Ren, S. Setty, A. J. Blumberg, and M. Walfish, "Verifying computations with state," in *Proceedings of the Twenty-Fourth ACM Symposium on Operating Systems Principles*. ACM, 2013, pp. 341–357.

[6] J.-G. Dumas and V. Zucca, "Prover efficient public verification of dense or sparse/structured matrix-vector multiplication," in *Information Security and Privacy*, ser. Lecture Notes in Computer Science, vol. 10343. Springer, 2017, pp. 115–134.

[7] I. Komargodski and O. Weinstein, "Proofs of useful work from arbitrary matrix multiplication (invited talk)," in *34th Annual European Symposium on Algorithms (ESA 2026)*, ser. LIPIcs, vol. 388, 2026, pp. 2:1–2:17.

[8] J. Alman, "Faster Walsh–Hadamard transform and matrix multiplication over finite fields using lookup tables," in *Symposium on Simplicity in Algorithms (SOSA)*. Society for Industrial and Applied Mathematics, 2023, pp. 137–144, full version arXiv:2211.04643v1, 9 November 2022; Section 1.1 and Theorem 1.2. [Online]. Available: https://arxiv.org/abs/2211.04643

[9] P. Bürgisser, M. Clausen, and M. A. Shokrollahi, *Algebraic Complexity Theory*, ser. Grundlehren der mathematischen Wissenschaften. Springer, 1997, vol. 315.

[10] V. Strassen, "Gaussian elimination is not optimal," *Numerische Mathematik*, vol. 13, no. 4, pp. 354–356, 1969.

[11] A. Cohen and G. Moshkovitz, "Structure vs. randomness for bilinear maps," *Discrete Analysis*, no. 12, pp. 1–21, 2022, an earlier version appeared in STOC 2021. [Online]. Available: https://arxiv.org/abs/2102.04657

[12] S. Kopparty, G. Moshkovitz, and J. Zuiddam, "Geometric rank of tensors and subrank of matrix multiplication," *Discrete Analysis*, no. 1, pp. 1–25, 2023, an earlier version appeared in CCC 2020. [Online]. Available: https://arxiv.org/abs/2002.09472

[13] D. Laksov and A. Thorup, "Counting matrices with coordinates in finite fields and of fixed rank," *Mathematica Scandinavica*, vol. 74, pp. 19–33, 1994.

[14] M. Karppa and P. Kaski, "Probabilistic tensors and opportunistic Boolean matrix multiplication," in *Proceedings of the Thirtieth Annual ACM-SIAM Symposium on Discrete Algorithms*. Society for Industrial and Applied Mathematics, 2019, pp. 496–515.

[15] M. Bläser, M. Christandl, and J. Zuiddam, "The border support rank of two-by-two matrix multiplication is seven," 2017, arXiv:1705.09652.

[16] W. Ogryczak and A. Tamir, "Minimizing the sum of the k largest functions in linear time," *Information Processing Letters*, vol. 85, no. 3, pp. 117–122, 2003.

[17] A. Federgruen and H. Groenevelt, "The greedy procedure for resource allocation problems: Necessary and sufficient conditions for optimality," *Operations Research*, vol. 34, no. 6, pp. 909–918, 1986.

[18] T. Wang, J. Blocki, N. Li, and S. Jha, "Locally differentially private protocols for frequency estimation," in *26th USENIX Security Symposium (USENIX Security 17)*. USENIX Association, 2017, pp. 729–745. [Online]. Available: https://www.usenix.org/conference/usenixsecurity17/technical-sessions/presentation/wang-tianhao

[19] J.-G. Dumas, P. Giorgi, and C. Pernet, "Dense linear algebra over word-size prime fields: the fflas and ffpack packages," *ACM Transactions on Mathematical Software*, vol. 35, no. 3, pp. 19:1–19:42, 2008.

[20] M. Christandl, V. Lysikov, and J. Zuiddam, "Weighted slice rank and a minimax correspondence to Strassen's spectra," *Journal de Mathématiques Pures et Appliquées*, vol. 172, pp. 299–329, 2023. [Online]. Available: https://arxiv.org/abs/2012.14412

[21] J. A. Blackard, "Covertype," 1998. [Online]. Available: https://archive.ics.uci.edu/dataset/31/covertype